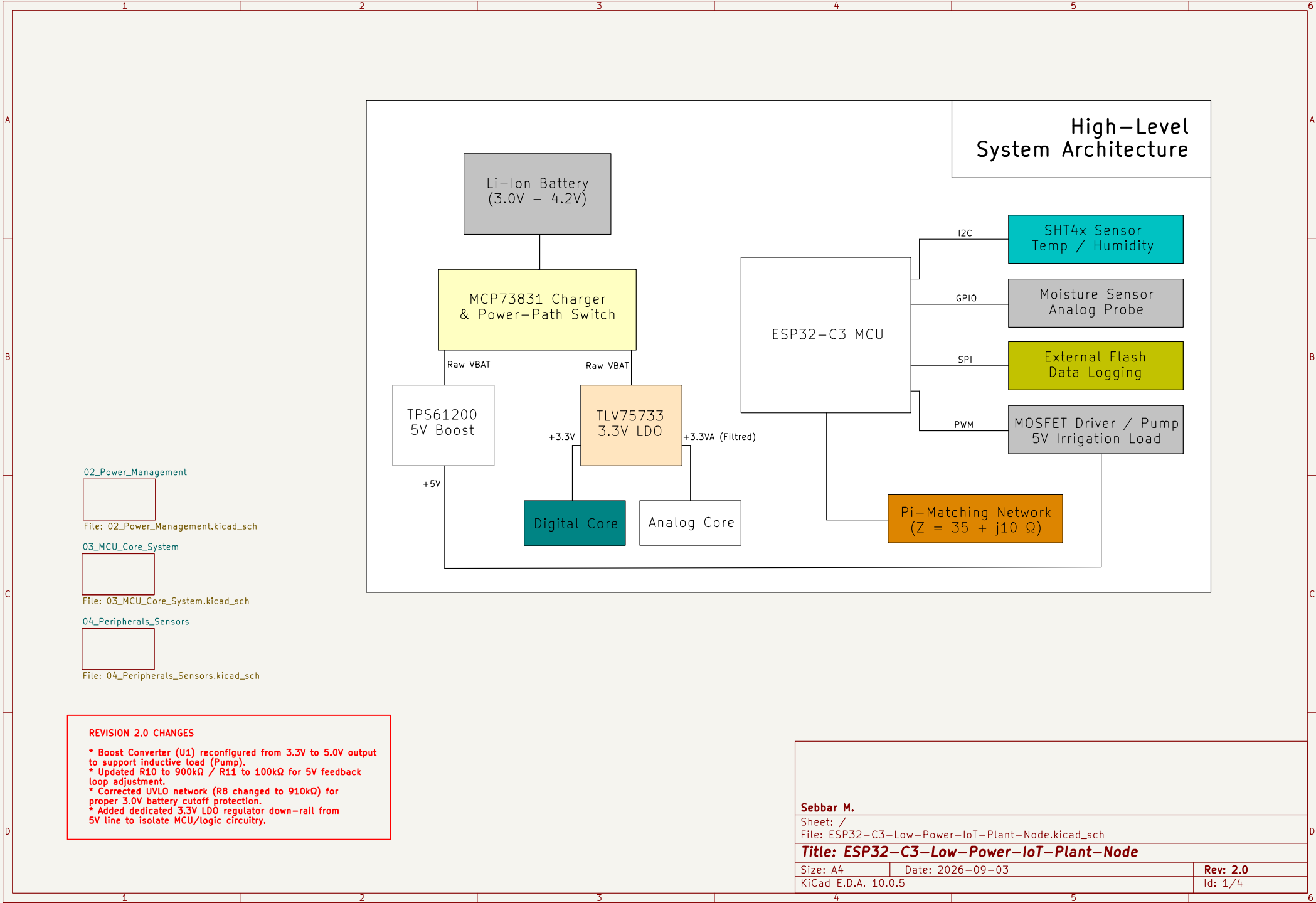




3.3V LDO REGULATOR

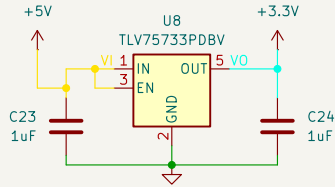
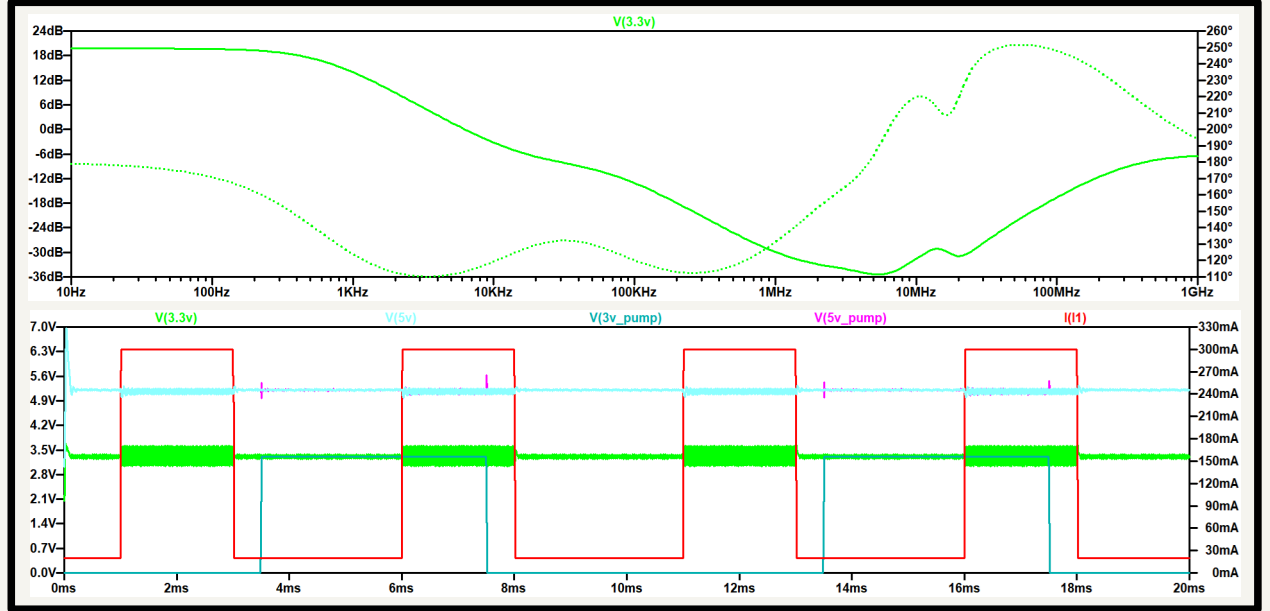


Table 4-1. Pin Functions

NAME	PIN	DBV	DYD	DRV	TYPE	DESCRIPTION
EN	3	3	4	I	I	Enable pin. Drive EN greater than V_{th} to turn on the regulator. Drive EN less than V_{LO} to place the LDO into shutdown mode.
GND	2	2	3	—	—	Ground pin.
IN	1	1	6	I	I	Input pin. A capacitor with a value of 1 μ F or larger is required from this pin to ground. ⁽¹⁾ See the Input and Output Capacitor Selection section for more information.
NC	4	4	2, 5	—	—	No internal connection.
OUT	5	5	1	O	O	Regulated output voltage pin. A capacitor with a value of 1 μ F or larger is required from this pin to ground. ⁽¹⁾ See the Input and Output Capacitor Selection section for more information.
Thermal pad	—	Pad	Pad	—	—	Connect the thermal pad to a large-area ground plane. The thermal pad is internally connected to GND.

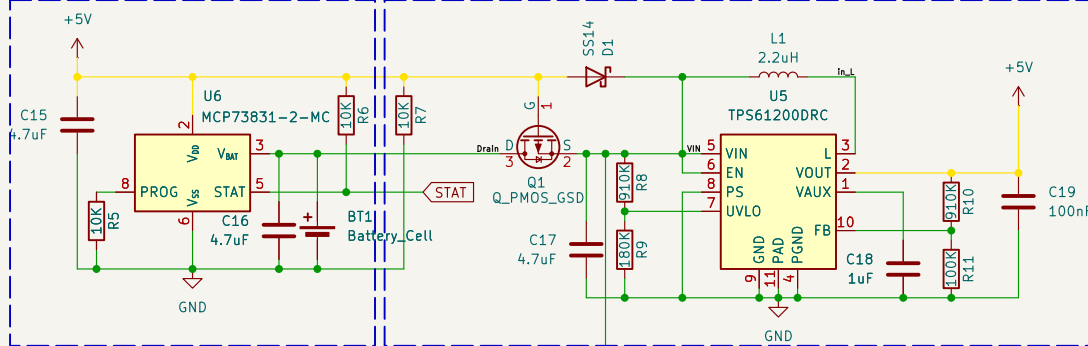
(1) Make sure the nominal input and output capacitance are greater than 0.47 μ F. Throughout this document the nominal derating on these capacitors is 50%. Make sure that the effective capacitance at the pin is greater than 0.47 μ F.

LTspice Simulation



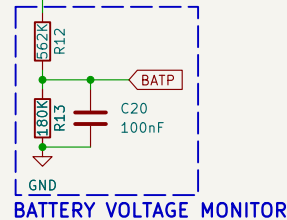
LI-ION BATTERY CHARGER

POWER PATH & 5V BOOST



Pin Functions

NAME	PIN	NO.	I/O	DESCRIPTION
EN	6	I	I	Enable input (High = enabled, Low = disabled). Do not leave floating.
Exposed thermal pad	—	—	—	Must be soldered to achieve appropriate power dissipation and mechanical reliability. Should be connected to PGND.
FB	10	I	I	Voltage feedback of adjustable versions, must be connected to V_{OUT} at fixed output voltage versions
GND	9	—	—	Control / logic ground
PGND	4	—	—	Power ground
PS	8	I	I	Enable/disable Power Save mode (High = disabled, Low = enabled). Do not leave floating.
L	3	I	I	Connection for Inductor
UVLO	7	I	I	Undervoltage lockout comparator input. Must be connected to VAUX if not used
VAUX	1	I/O	I/O	Supply voltage for control stage
VIN	5	I	I	Boost converter input voltage
VOUT	2	O	O	Boost converter output



BATTERY VOLTAGE MONITOR

Sebbar M.

Sheet: /02_Power_Management/
File: 02_Power_Management.kicad_sch

Title: ESP32-C3-Low-Power-IoT-Plant-Node

Size: A4 Date: 2026-09-03

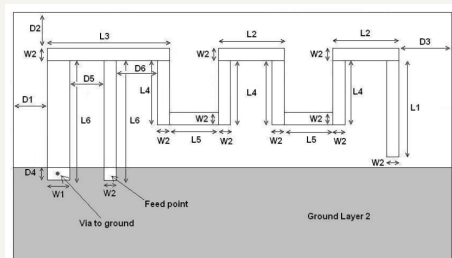
KiCad E.D.A. 10.0.5

Rev: 2.0

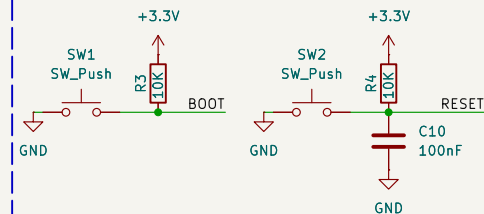
Id: 2/4

L1	3.94 mm
L2	2.70 mm
L3	5.00 mm
L4	2.64 mm
L5	2.00 mm
L6	4.90 mm
W1	0.90 mm
W2	0.50 mm
D1	0.50 mm
D2	0.30 mm
D3	0.30 mm
D4	0.50 mm
D5	1.40 mm
D6	1.70 mm

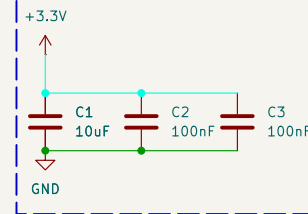
Antenna Dimensions



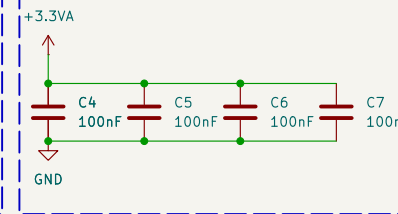
RESET + BOOT CIRCUIT



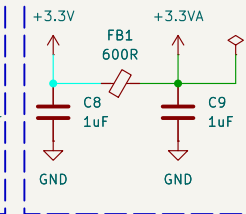
DIGITAL DECOUPLING



ANALOG DECOUPLING



ANALOG FILTERING



ANTENNA CIRCUIT

Gain in XY plane	4.5 dB
Gain in XZ plane	5.3 dB
Gain in YZ plane	5.3 dB
Gain in XY plane, connected to laptop	3.3 dB
LOS range	240 m
Antenna size	15.2 x 5.7 mm

Table 6: IFA Properties (Measured on CC2511 Dongle Reference Design)

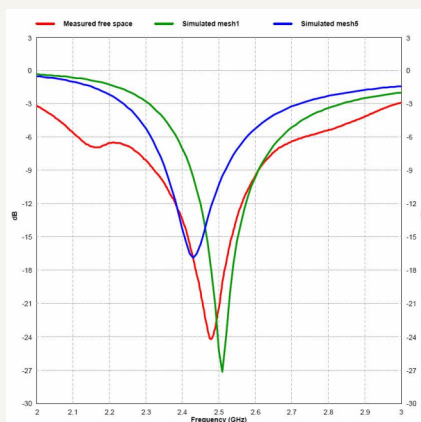
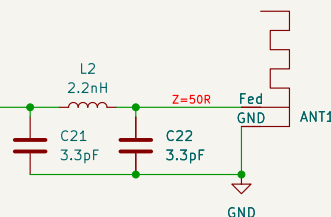
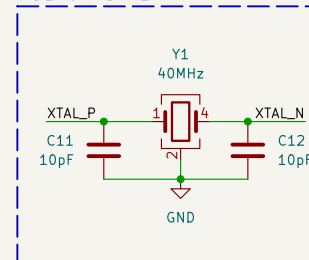
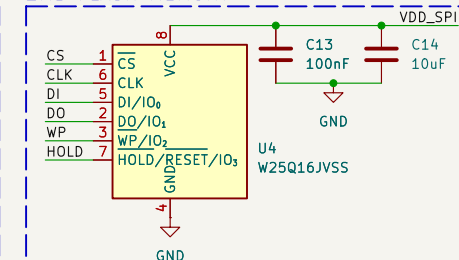


Figure 2: Comparison of Simulation and Measurements Results

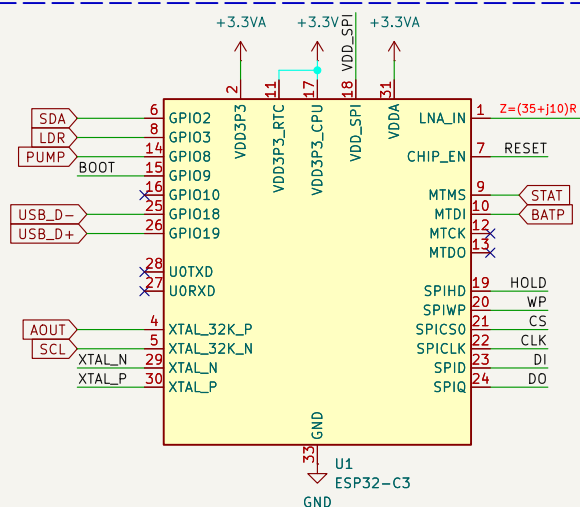
HSE CRYSTAL



2MB FLASH MEMORY



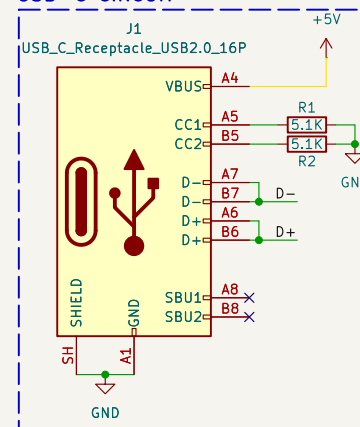
ESP32-C3 MCU



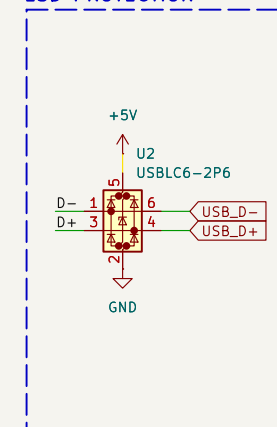
CHECK OUT THE FULL ANTENNA AN043:

<https://www.ti.com/lit/an/swra117d/swra117d.pdf>

USB-C CIRCUIT



ESD PROTECTION



Sebbar M.

Sheet: /03_MCU_Core_System/
File: 03_MCU_Core_System.kicad_sch

Title: ESP32-C3-Low-Power-IoT-Plant-Node

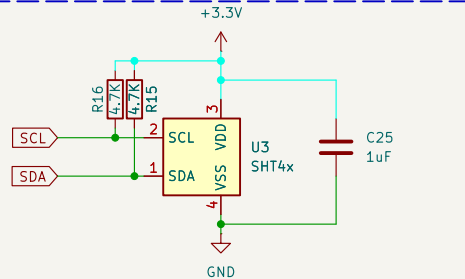
Size: A4
KiCad E.D.A. 10.0.5

Date: 2026-09-03

Rev: 2.0

Id: 3/4

TEMP/HUM SENSOR



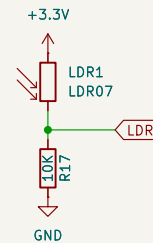
```
// Address & Command
#define SHT4X_ADDR 0x44
#define SHT4X_CMD 0xFD

// Read Function
void read_sht4x(float *temp, float *humidity) {
  i2c_write(SHT4X_ADDR, SHT4X_CMD);
  delay_ms(10);
  uint8_t rx[6] = i2c_read(SHT4X_ADDR, 6);

  uint16_t t_ticks = (rx[0] << 8) | rx[1];
  uint16_t rh_ticks = (rx[3] << 8) | rx[4];

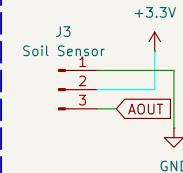
  *temp = -45 + 175 * ((float)t_ticks / 65535);
  *humidity = -6 + 125 * ((float)rh_ticks / 65535);
}
```

LDR



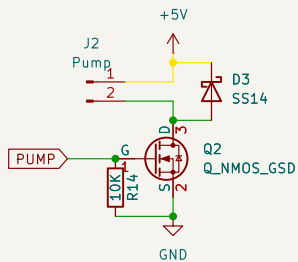
```
// Read Function (0-4095: ~164 Dark, ~3931 Bright)
uint16_t read_ldr(void) {
  return analogRead(PIN_LDR);
}
```

SOIL SENSOR



```
// Read Function (0-4095: ~819 Dry, ~3604 Saturated)
uint16_t read_soil(void) {
  return analogRead(PIN_SOIL);
}
```

PUMP CIRCUIT



```
// Control Functions (1kHz PWM / 60s Auto-Cutoff)
void pump_on(uint8_t speed_pwm) {
  analogWrite(PIN_PUMP, speed_pwm);
  start_timer_60s_limit();
}

void pump_off(void) {
  analogWrite(PIN_PUMP, 0);
  stop_timer();
}
```

Pin Definition

```
#define PIN_SDA 2
#define PIN_SCL 5
#define PIN_LDR 3
#define PIN_SOIL 4
#define PIN_PUMP 8
```



Sebbar M.

Sheet: /04_Peripherals_Sensors/
File: 04_Peripherals_Sensors.kicad_sch

Title: ESP32-C3-Low-Power-IoT-Plant-Node

Size: A4 Date: 2026-09-03

KiCad E.D.A. 10.0.5

Rev: 2.0

Id: 4/4